

ОБРАЗОВАНИЕ В ПЕДИАТРИЧЕСКОМ УНИВЕРСИТЕТЕ

# Эпителиальные ткани



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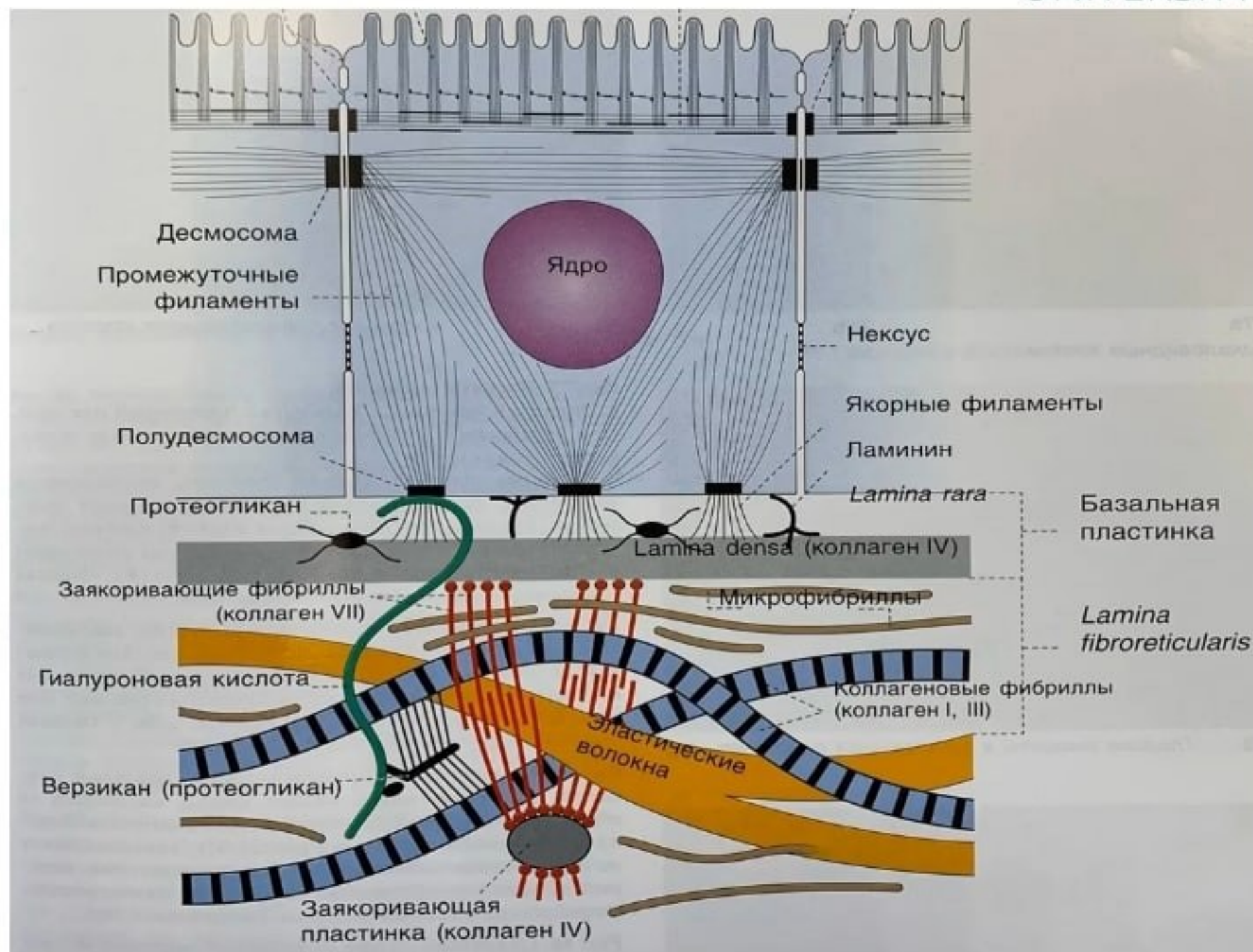
**Кожухарь Владимир  
Гарибальдиевич**

Заведующий кафедрой гистологии и  
эмбриологии им.проф. А.Г. Кнорре  
СПбГПМУ,  
к.м.н., доцент



## Схема строения базальной мембраны

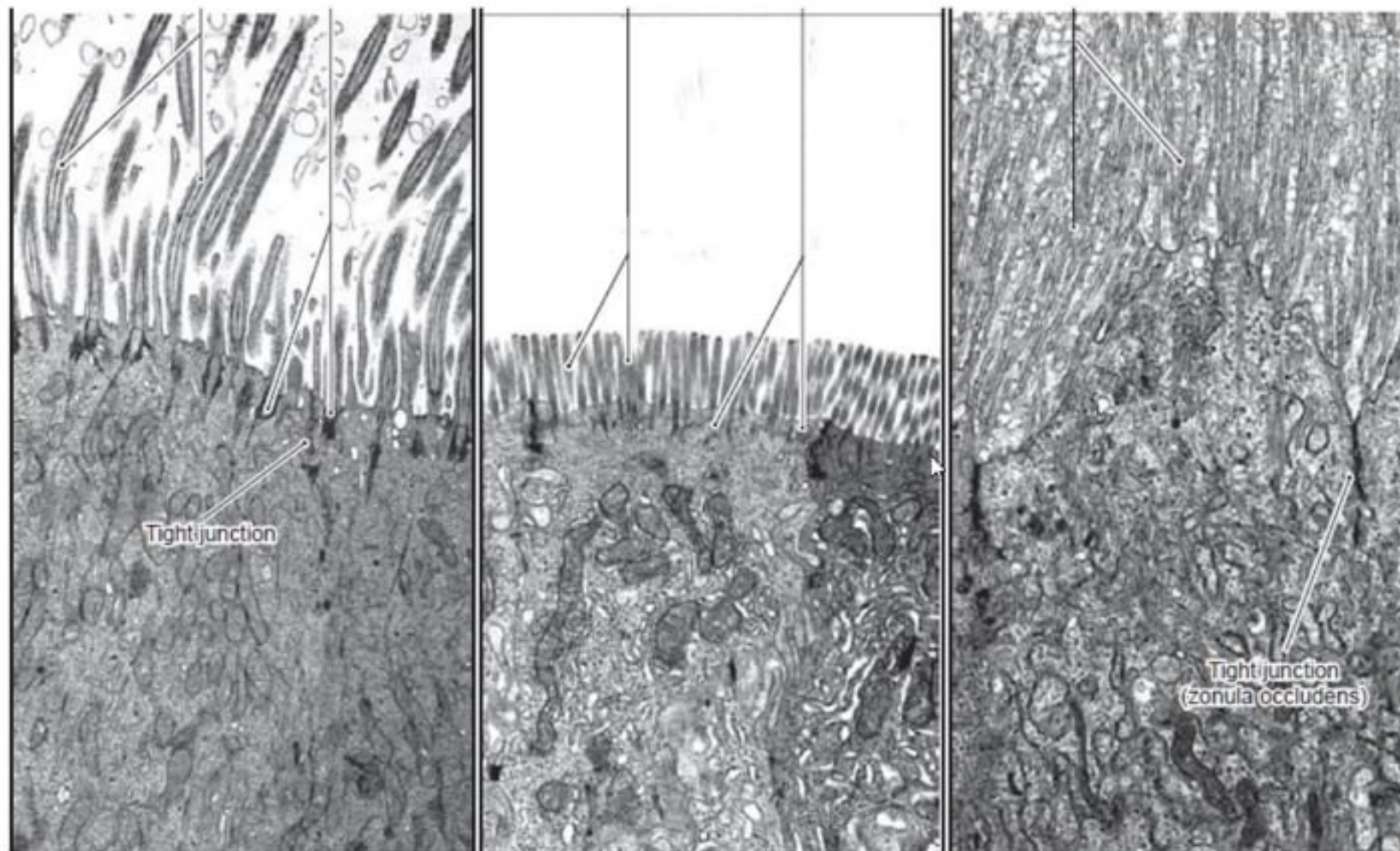
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Атлас гистологии под ред. У. Вельша, 2011

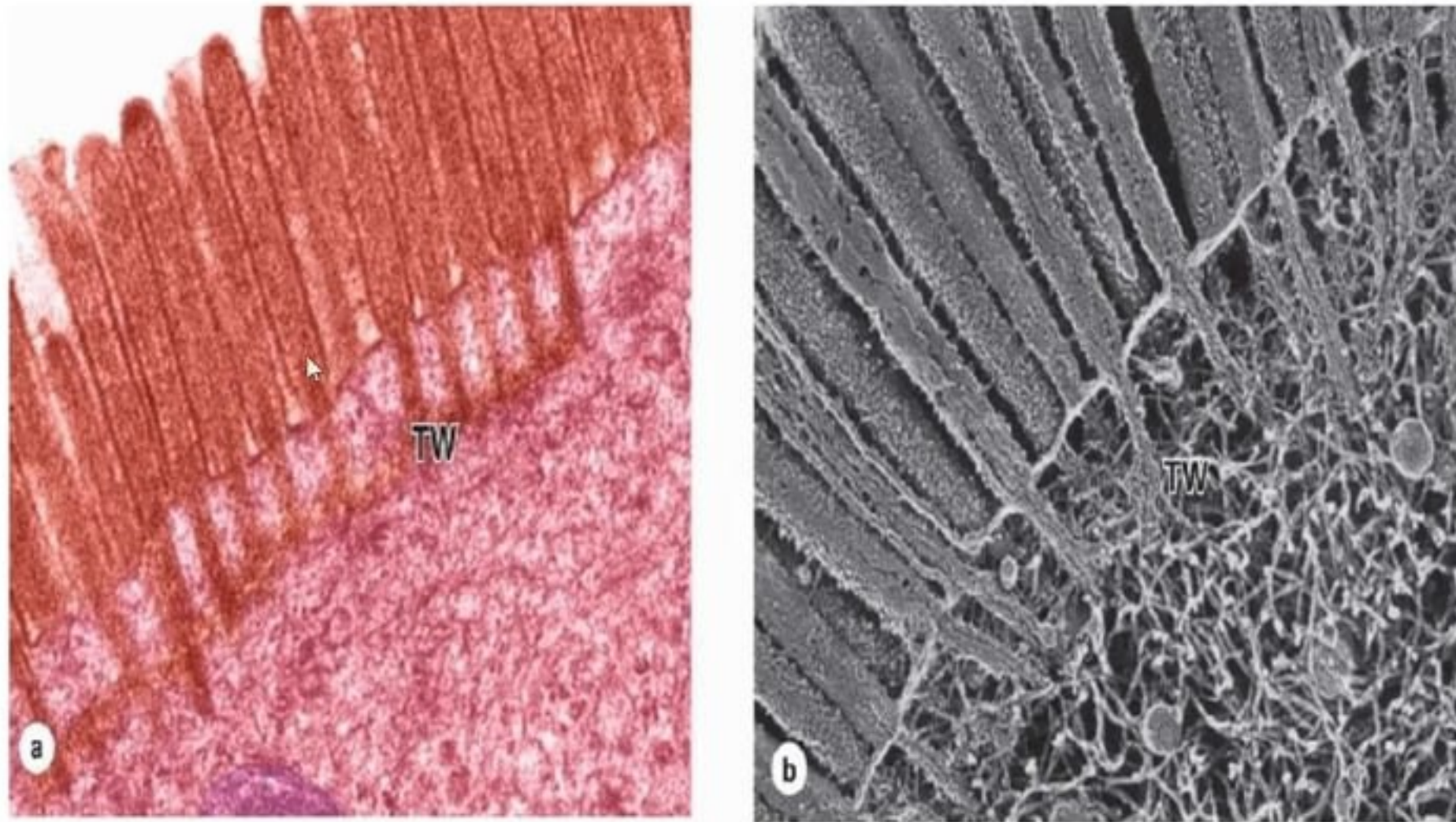


## Мерцательные реснички, микроворсинки щеточной каемки, стереоцилии



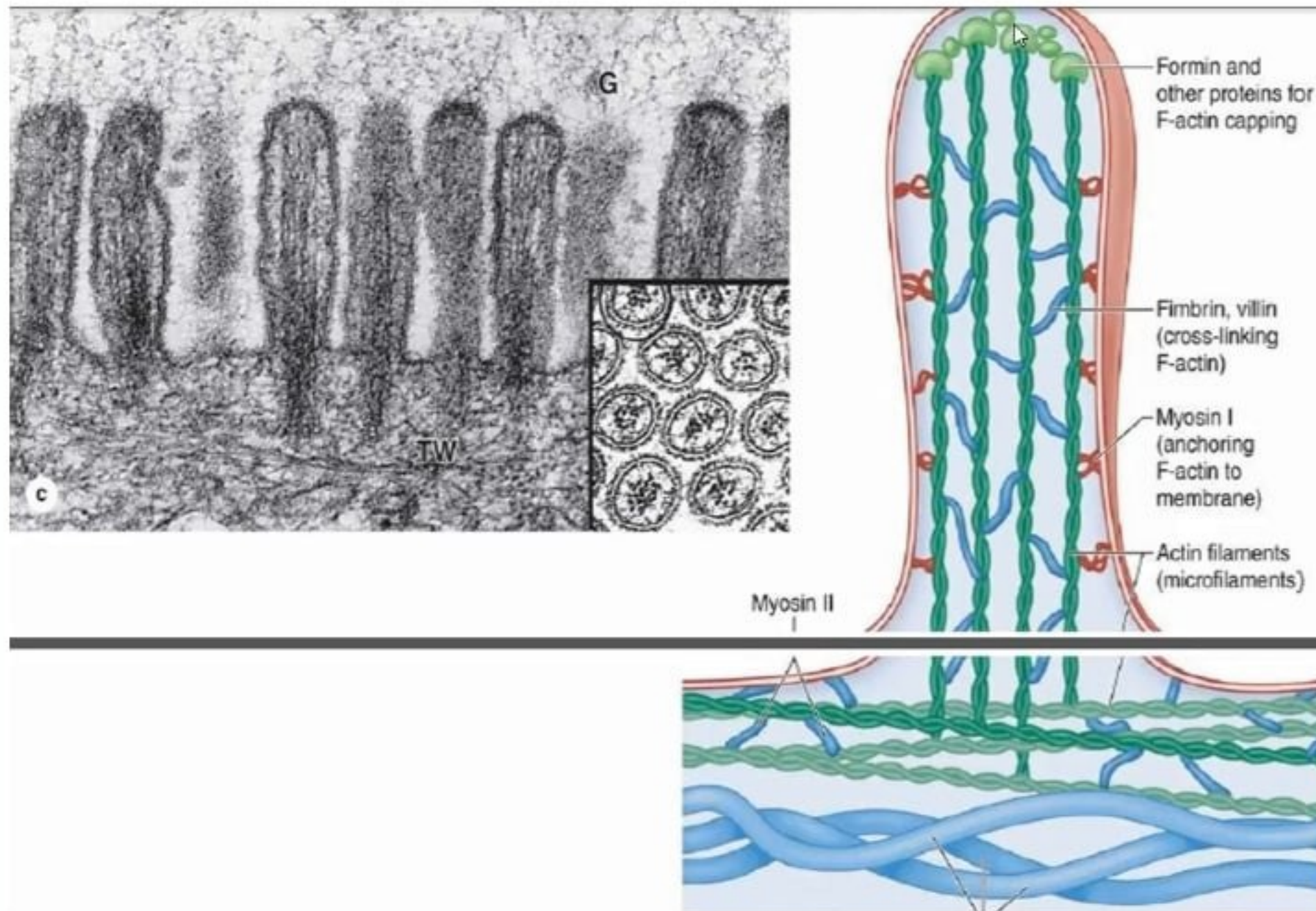
Atlas of Histology with Functional and Clinical Correlations, D.Cui, 2011







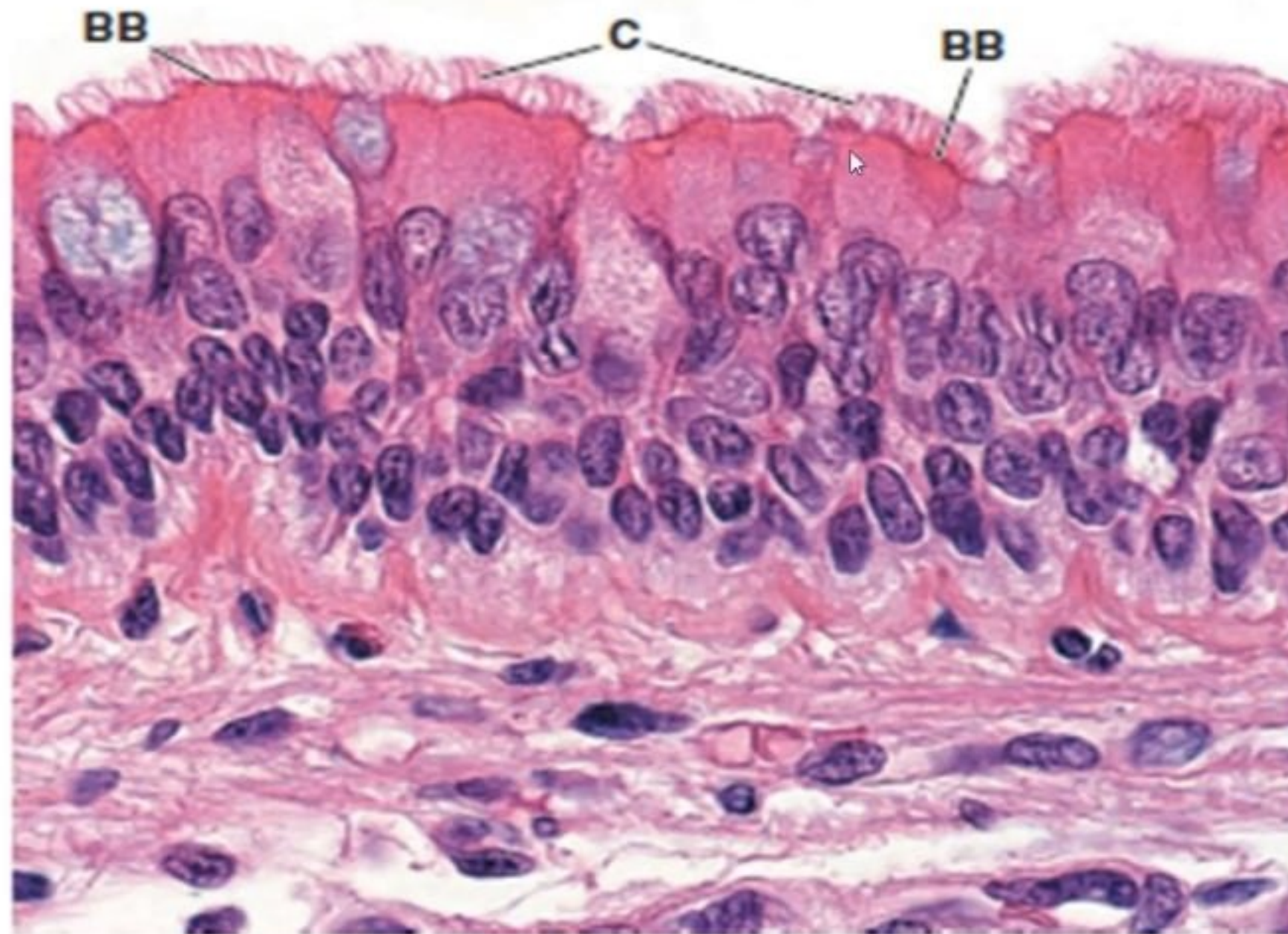
## Микроворсинки всасывающей (щеточной) каемки





Однослойный многорядный реснитчатый эпителий  
трахеи

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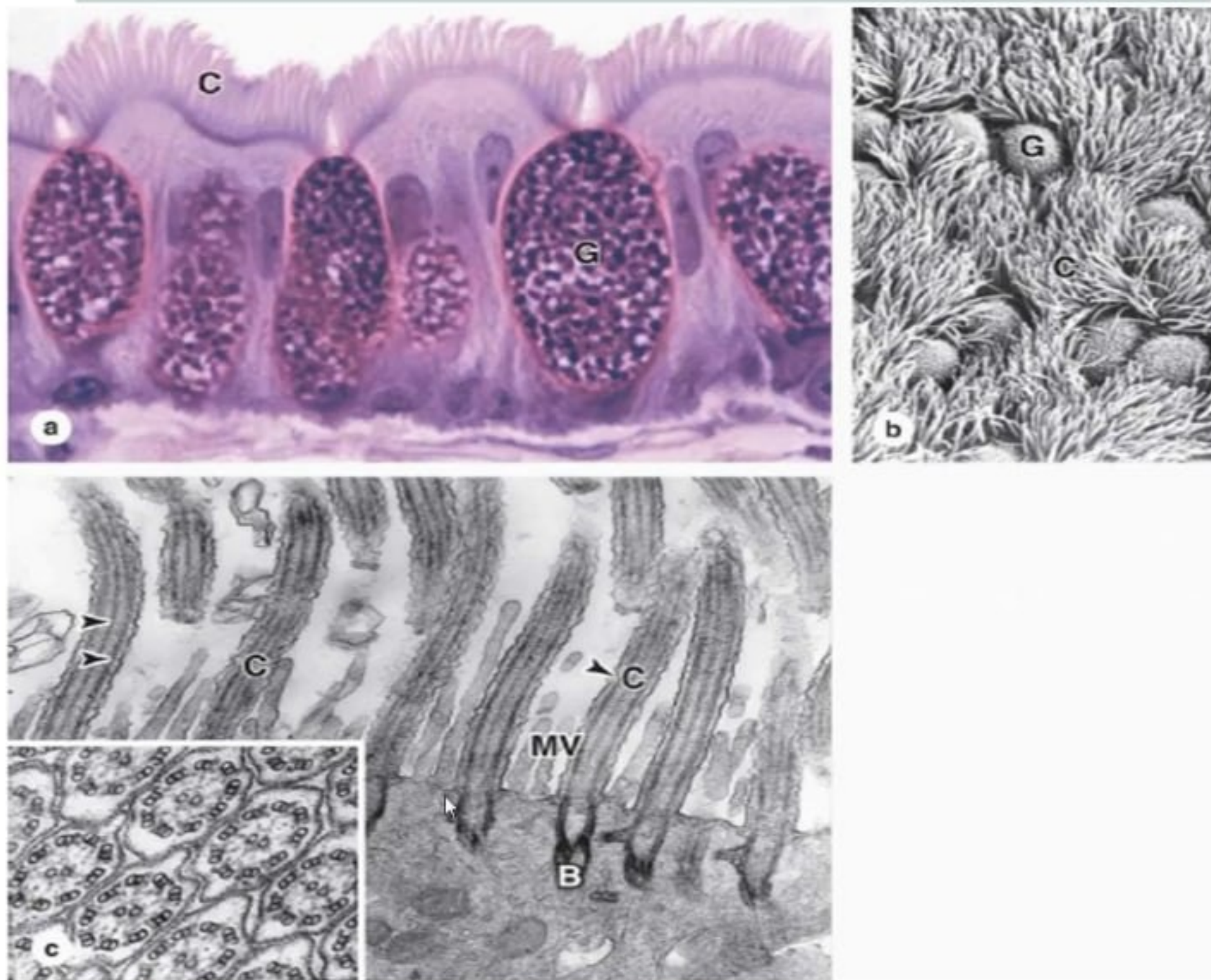


Junqueira's Basic Histology, A.L. Mescher, 13th ed., 2013



# Мерцательные реснички клеток эпителия воздухоносных путей

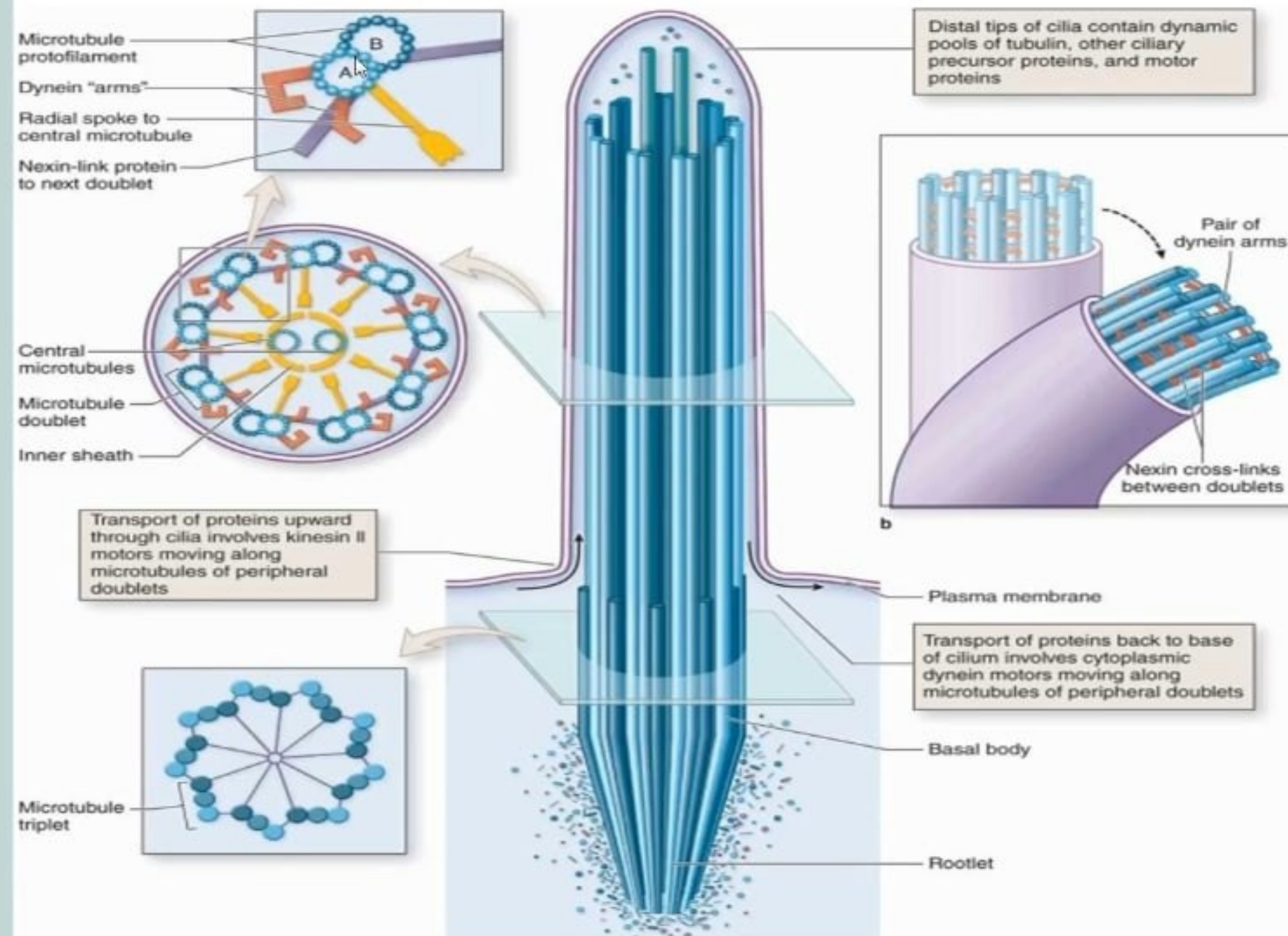
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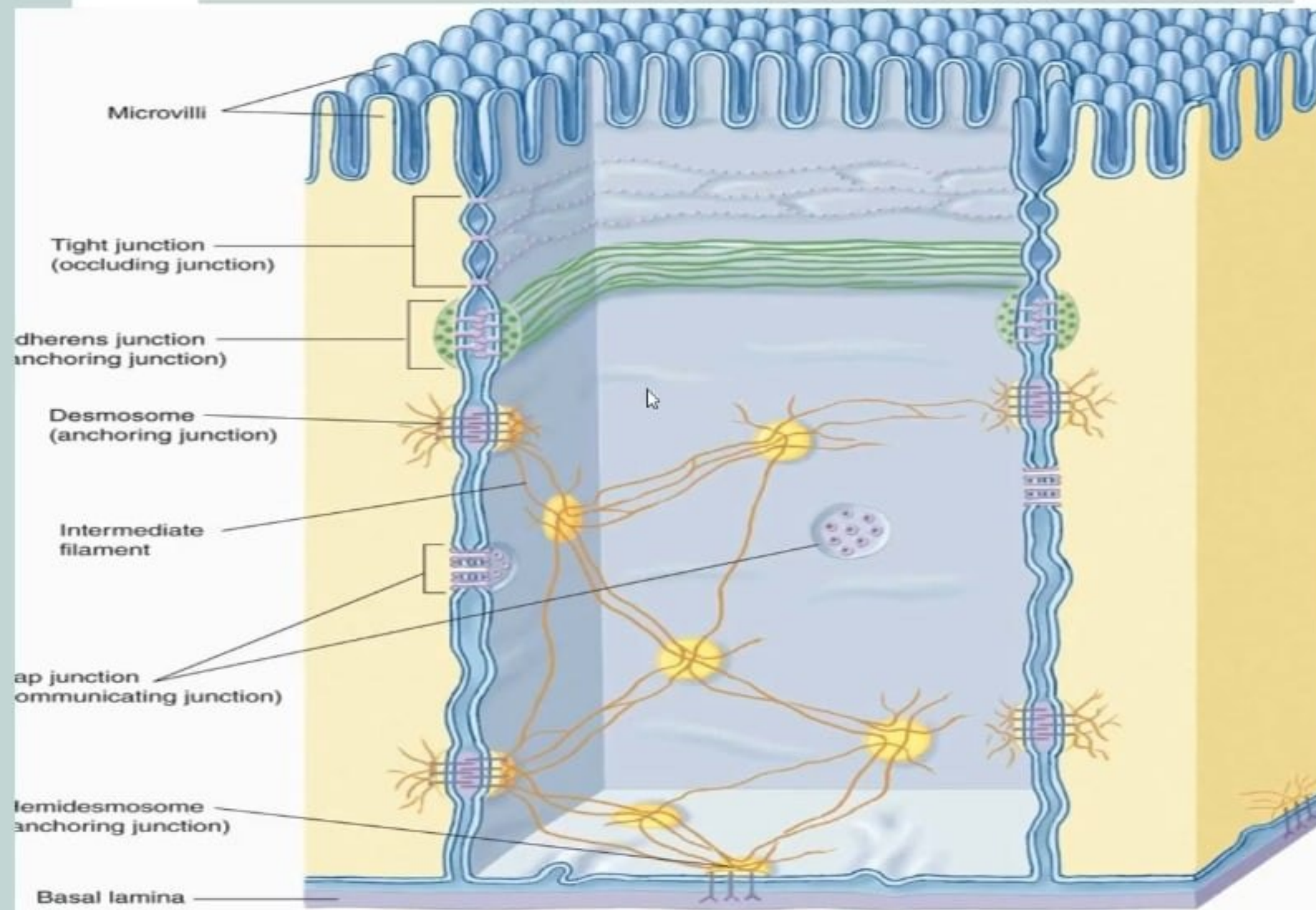
Junqueira's Basic Histology, A.L. Mescher, 13th ed., 2013



## Схема строения мерцательной реснички



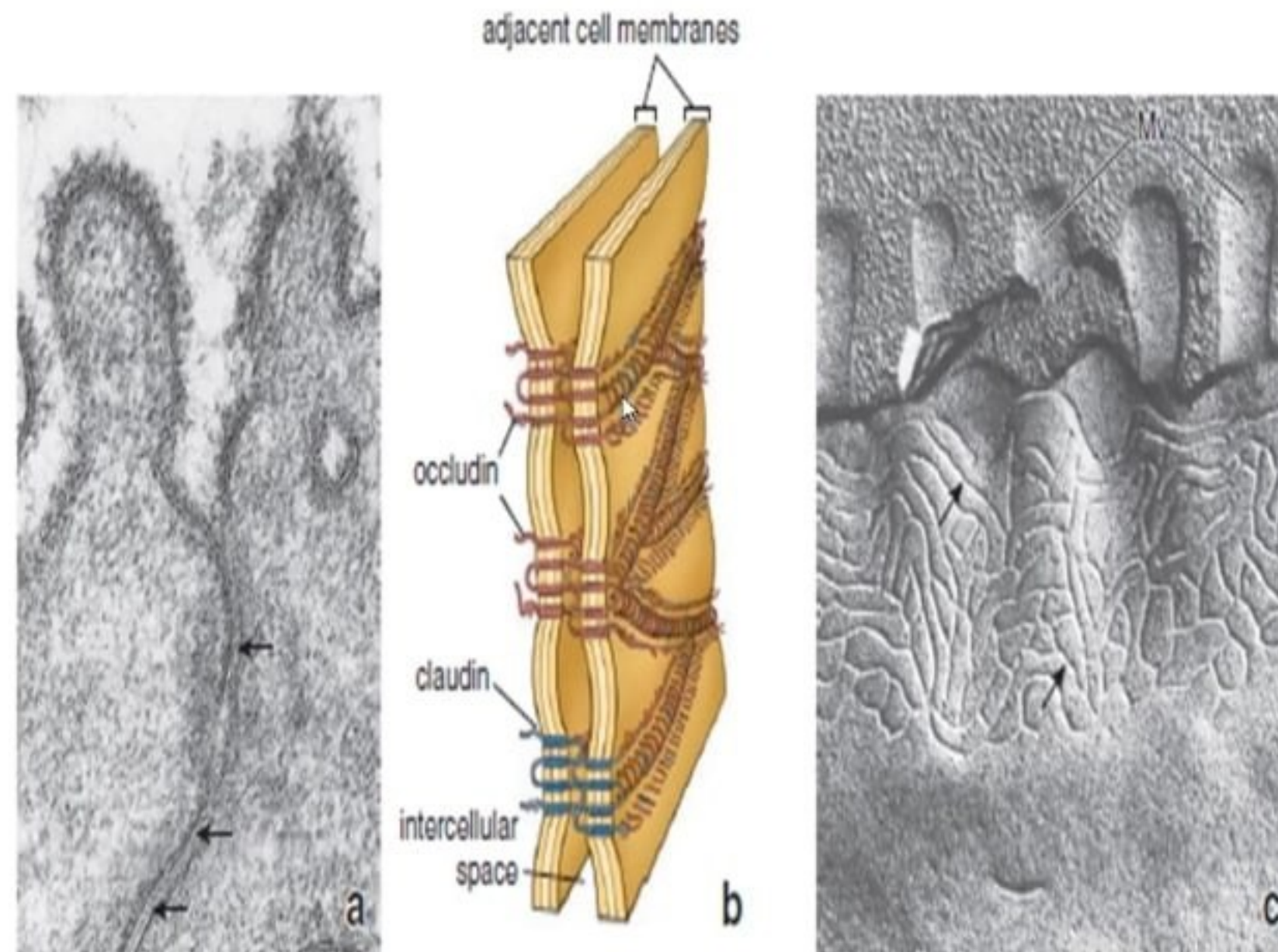






## Схема строения плотного соединения (zonula occludens)

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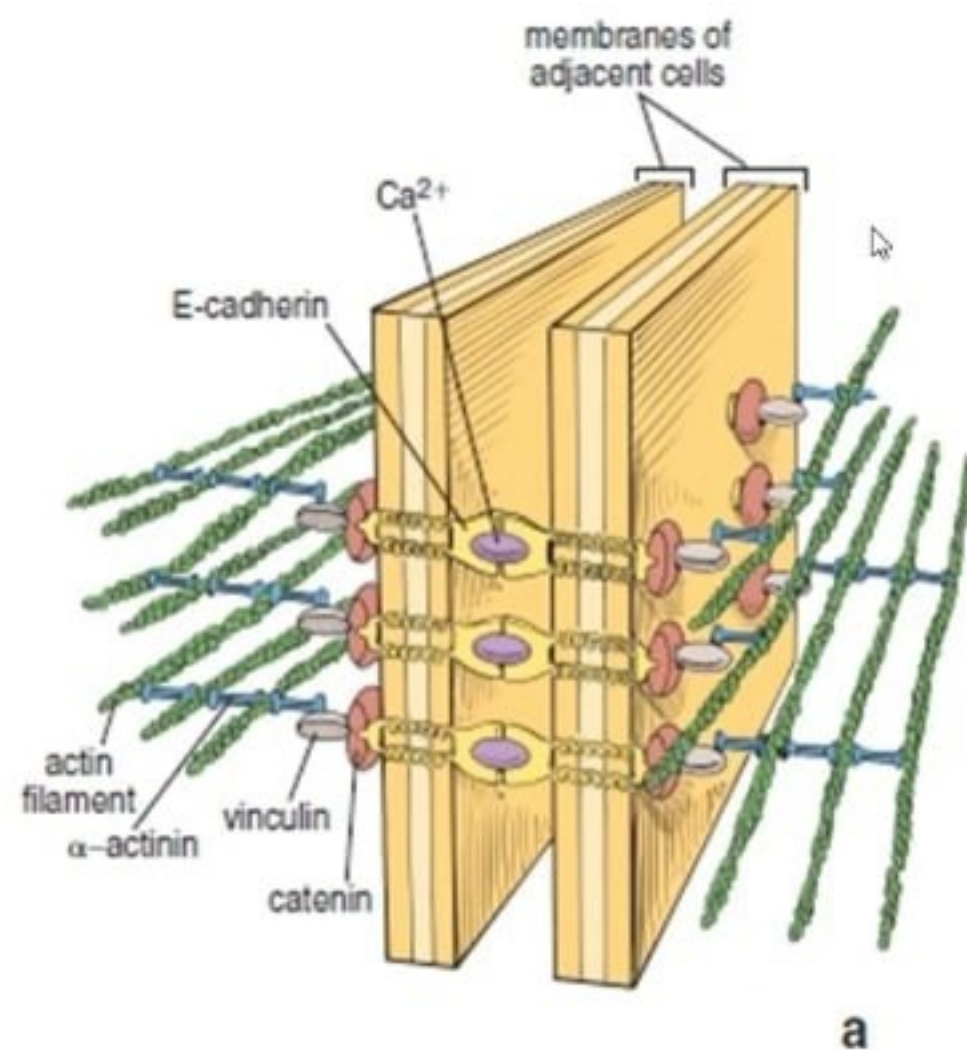


Ross M. H. , Pawlina W. Histology, 6th ed., 2011



## Схема строения опоясывающей десмосомы (zonula adhereus)

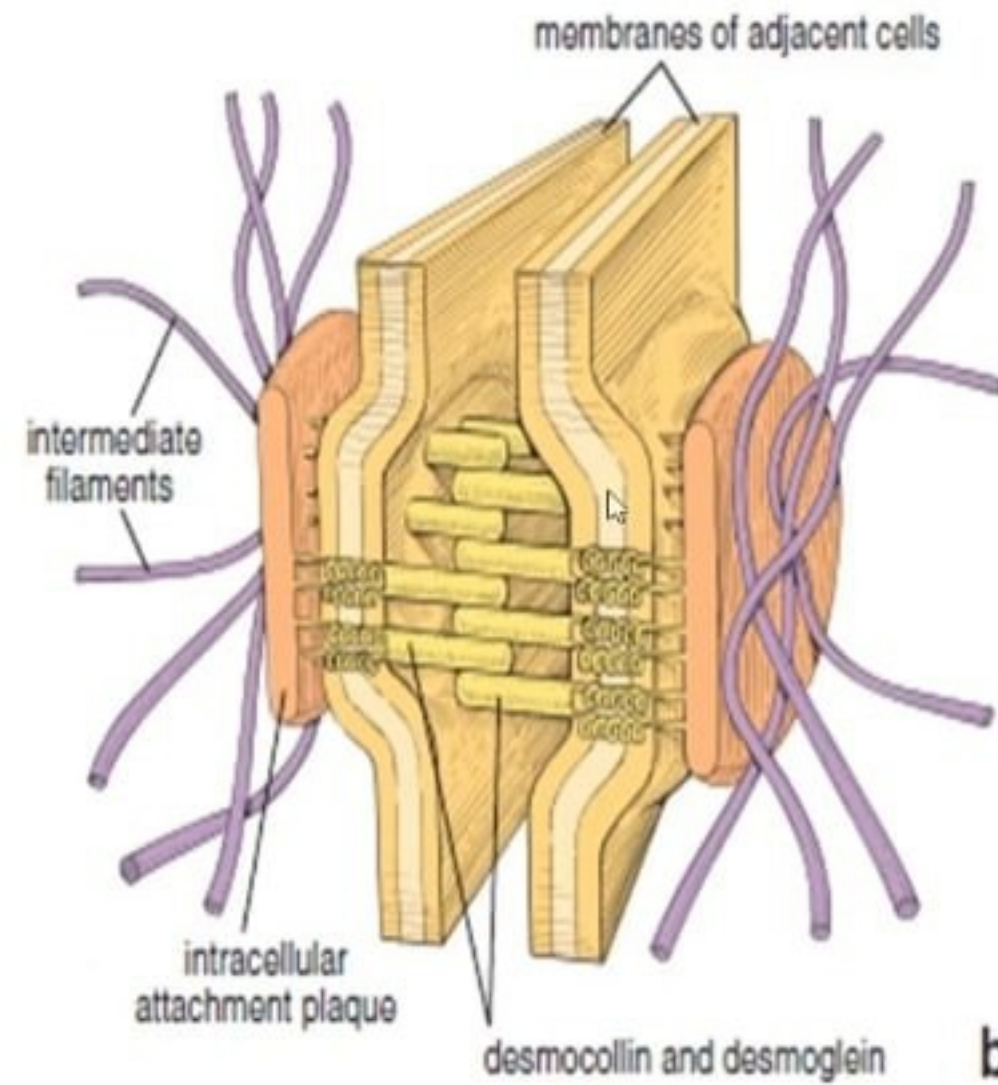
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Ross M. H. , Pawlina W. Histology, 6th ed., 2011

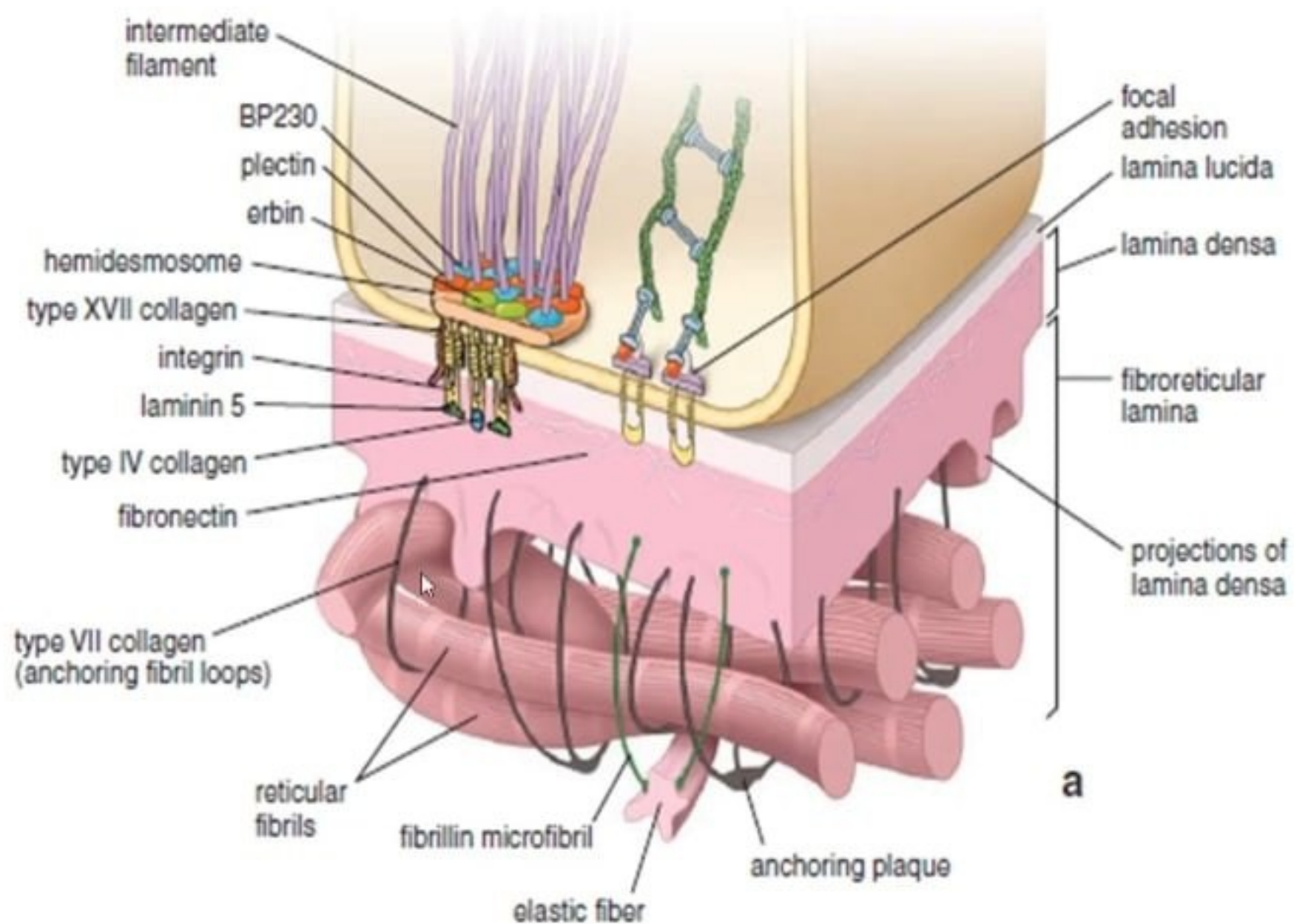
## Схема строения десмосомы (makula adhereus)

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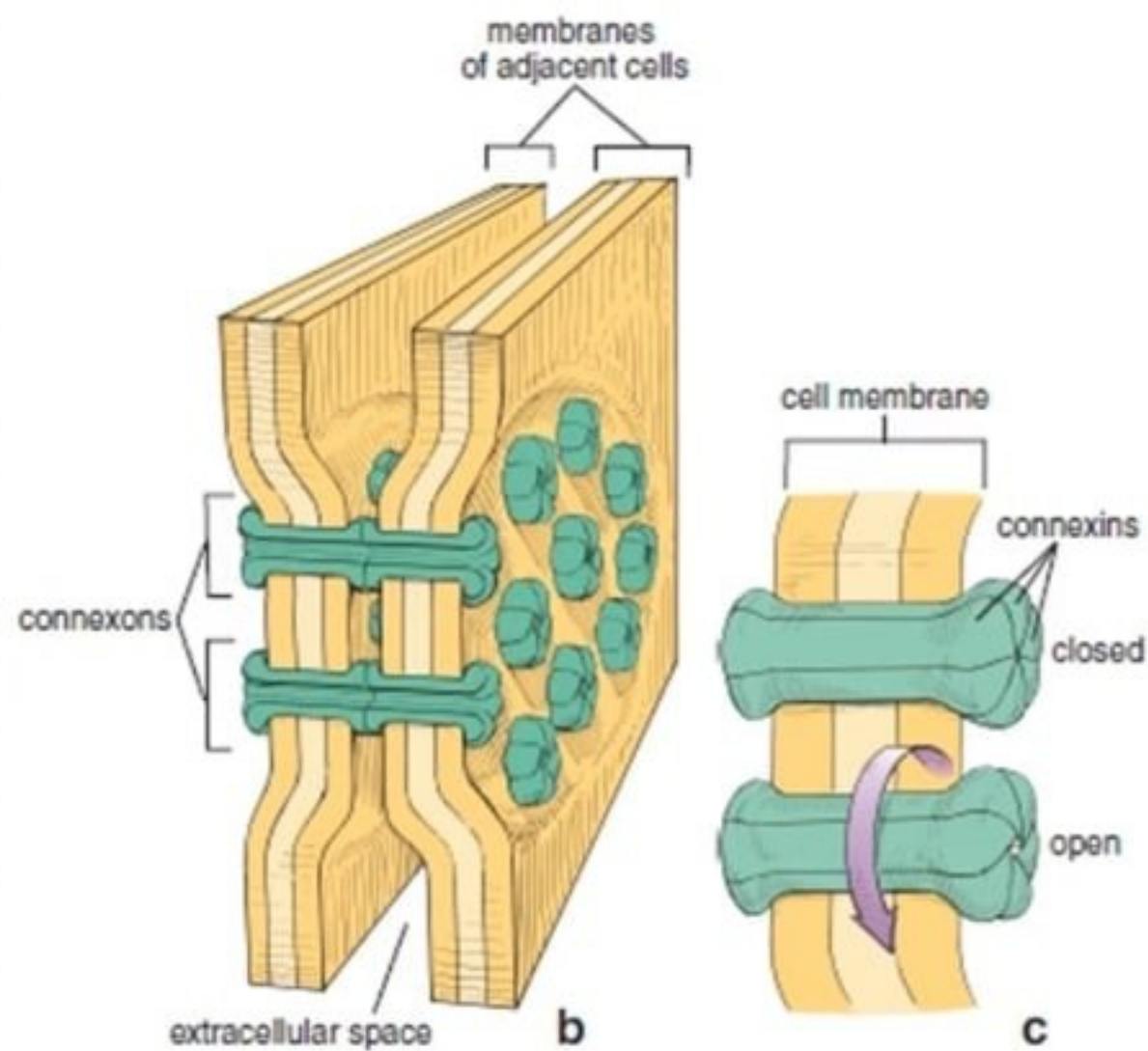


Ross M. H. , Pawlina W. Histology, 6th ed., 2011





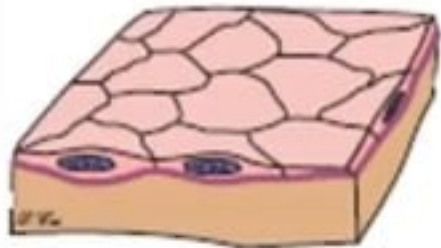
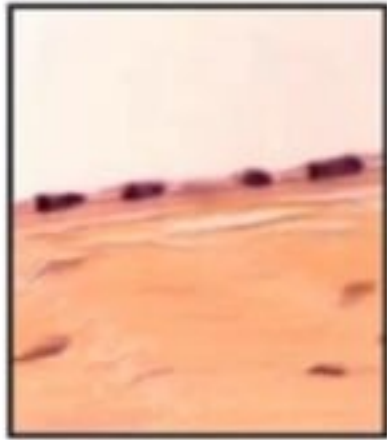
## Схема строения щелевого соединения (нексус)



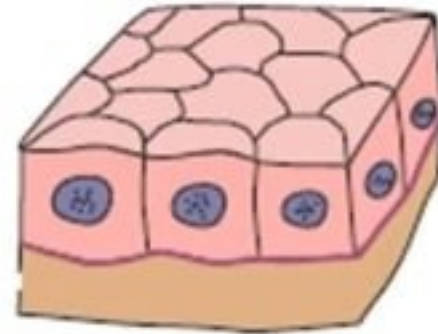
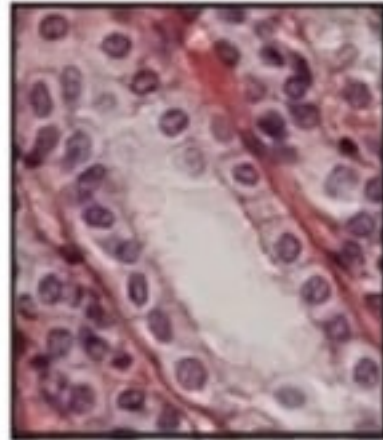
Ross M. H. , Pawlina W. Histology, 6th ed., 2011



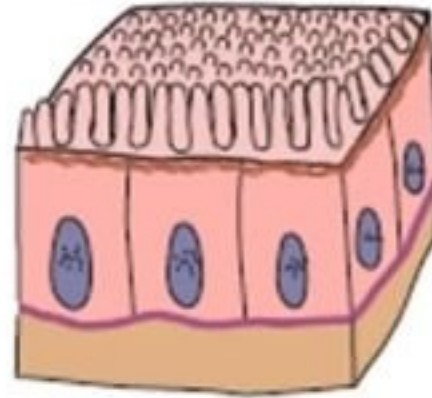
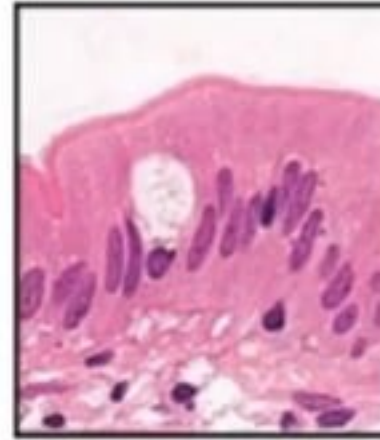
Squamous  
epithelium



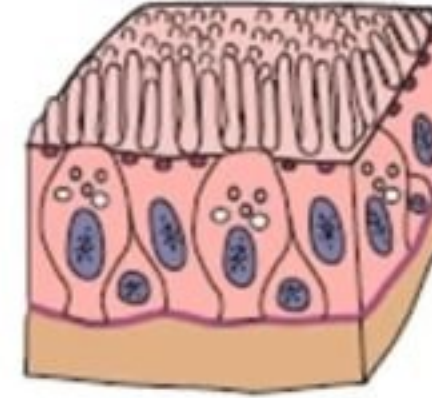
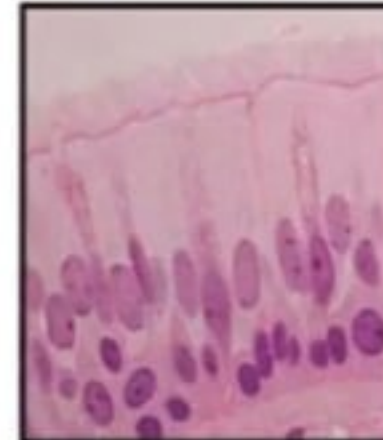
Cuboidal  
epithelium

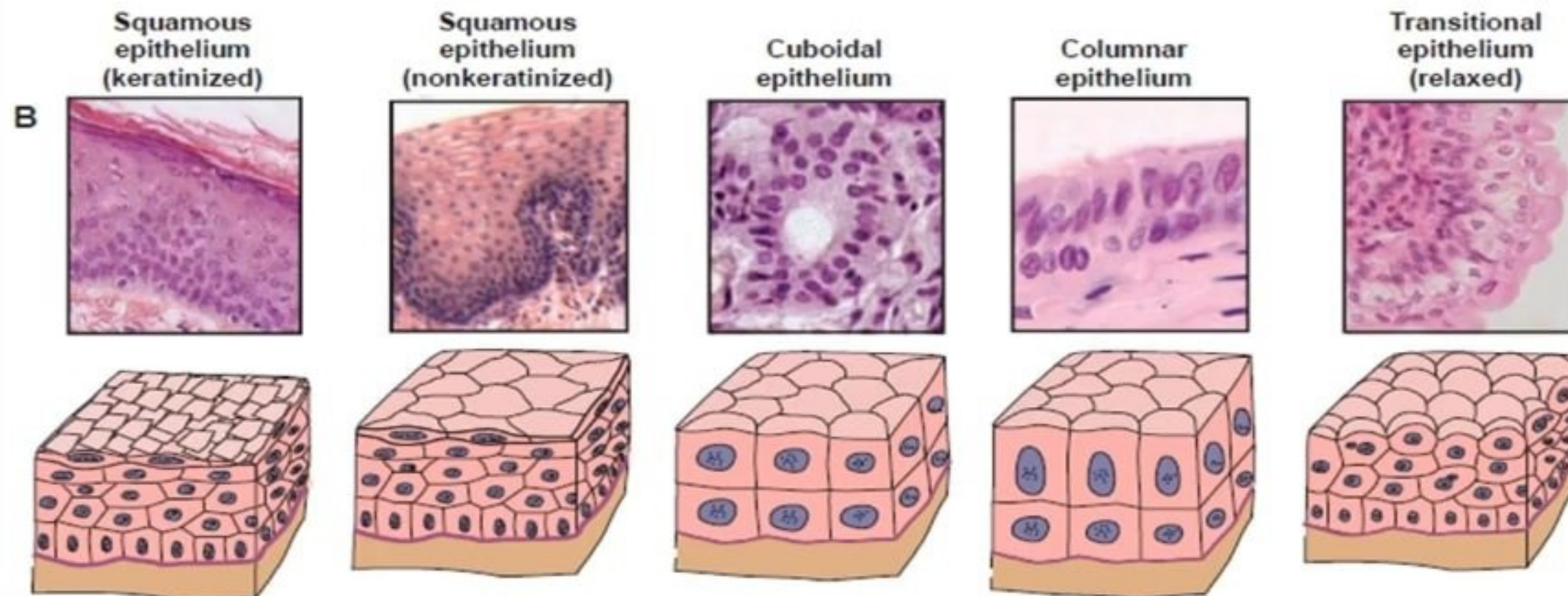


Columnar  
epithelium



Pseudostratified  
columnar epithelium

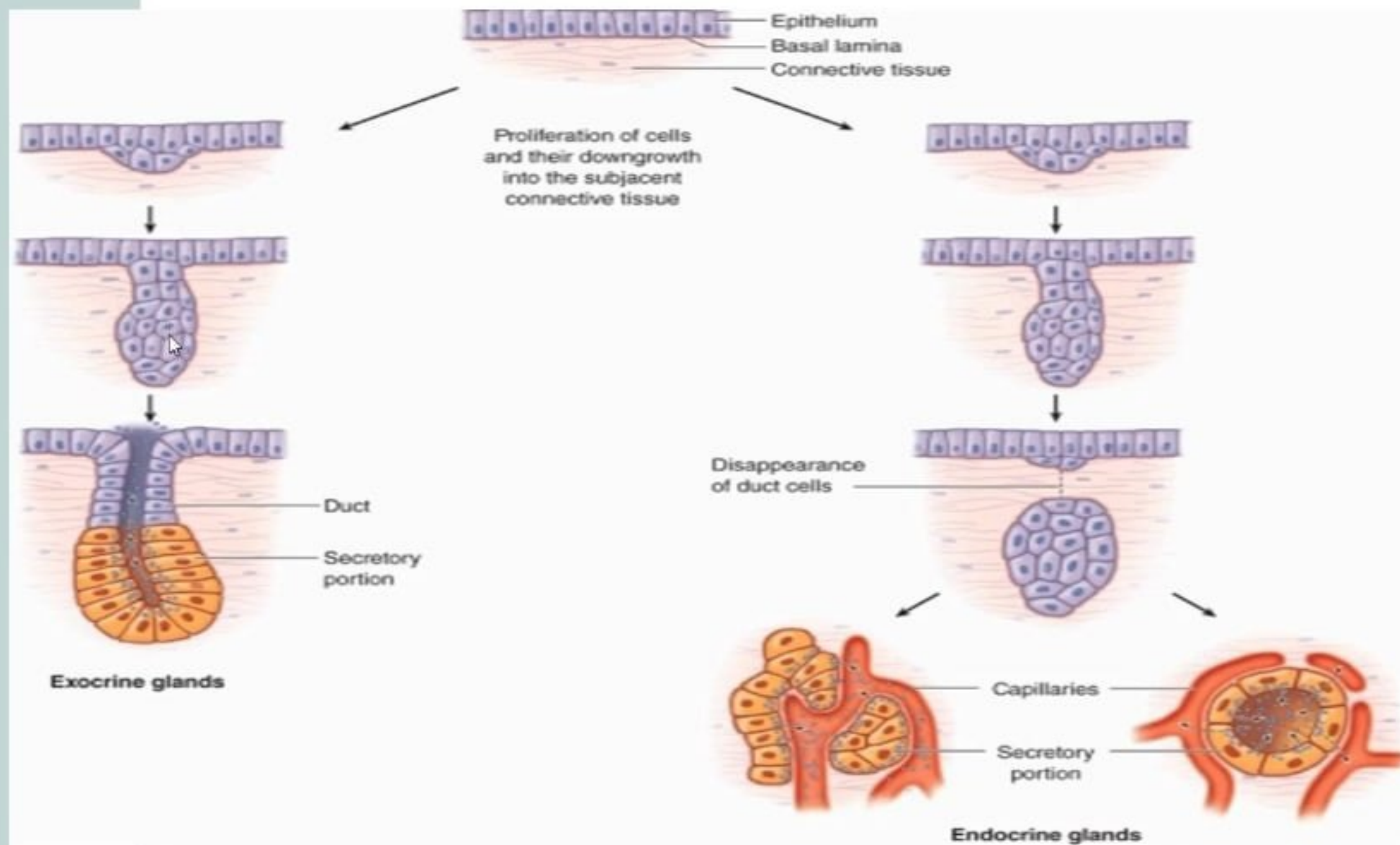




Atlas of Histology with Functional and Clinical Correlations, D.Cui, 2011



## Схема развития экзо- и эндокринных желез



# Классификация экзокринных желез (простые железы)

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|               | Classification                                                                                                   | Typical Location                                | Features                                                                                                                                         |
|---------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Simple Glands | Simple tubular<br>            | Large intestine: intestinal glands of the colon | Secretory portion of the gland is a straight tube formed by the secretory cells (goblet cells)                                                   |
|               | Simple coiled tubular<br>     | Skin: eccrine sweat gland                       | Coiled tubular structure is composed of the secretory portion located deep in the dermis                                                         |
|               | Simple branched tubular<br> | Stomach: mucus-secreting glands of the pylorus  | Branched tubular glands with wide secretory portion are formed by the secretory cells and produce a viscous mucous secretion                     |
|               | Simple acinar<br>           | Urethra: paraurethral and periurethral glands   | Simple acinar glands develop as an outpouching of the transitional epithelium and are formed by a single layer of secretory cells                |
|               | Branched acinar<br>         | Stomach: mucus-secreting glands of cardia       | Branched acinar glands with secretory portions are formed by mucus-secreting cells; the short, single-duct portion opens directly into the lumen |

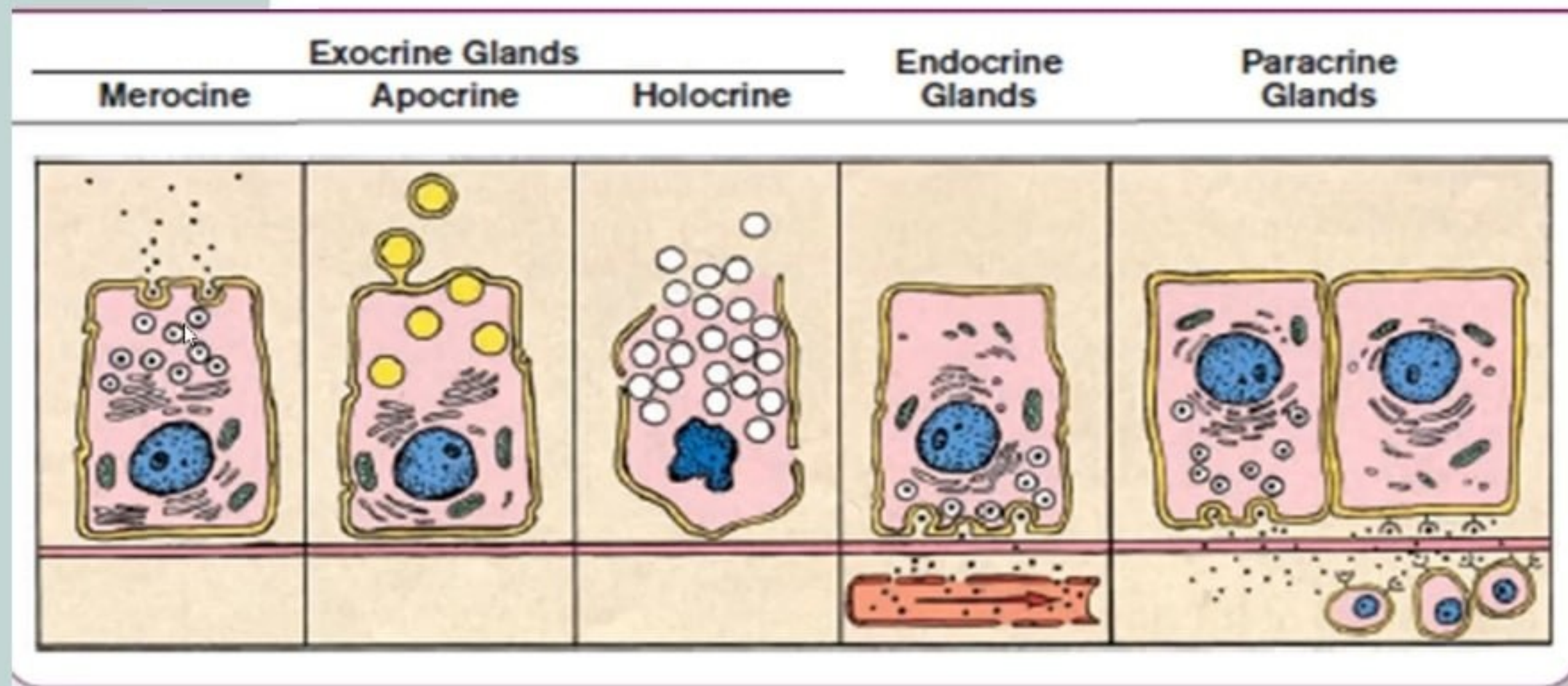
Ross M. H. , Pawlina W. Histology, 6th ed., 2011



# Классификация экзокринных желез (сложные железы)

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|                 |                       |                                                                                      |                                                             |                                                                                                                                                      |
|-----------------|-----------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compound Glands | Compound tubular      |    | Duodenum: submucosal glands of Brunner                      | Compound tubular glands with coiled secretory portions are located deep in the submucosa of the duodenum                                             |
|                 | Compound acinar       |   | Pancreas: excretory portion                                 | Compound acinar glands with alveolar-shaped secretory units are formed by pyramid-shaped serous-secreting cells                                      |
|                 | Compound tubuloacinar |  | Submandibular salivary gland, mammary gland, lacrimal gland | Compound tubuloacinar glands can have both mucous branched tubular and serous branched acinar secretory units; they have serous end-caps (demilunes) |



Ross M. H. , Pawlina W. Histology, 6th ed., 2011



